

Radio Astronomy Imaging

1. Description

In radio astronomy, an array of antennas (classically dishes) pick up signals in a frequency range between 30 Hz – 30 GHz, and correlate them. During imaging, the correlations are converted into a spatial power spectrum: a map of the sky. This is very similar to power spectrum estimating which we have seen in class, but now in two dimensions. Also, the spatial samples (distances between antennas) are typically not uniform.

A recently developed instrument is LOFAR: a distributed radio telescope observing between 30 – 240 Hz, consisting of some 50 stations distributed over Europe. Each station has some 200 antennas. The central stations are close to each other and form the LOFAR Core (located in Exloo, The Netherlands).

Already the antennas in a single station can be used to create a sky image, albeit at a very coarse resolution and based on only a few seconds of data.

How can you apply the various power spectrum estimation techniques (periodogram, MVDR, perhaps MUSIC) to create an image? Are other techniques relevant as well? E.g., consider the Adaptive Angular Response (AAR), which is

Slides

2. Dataset

LOFAR is a large radio telescope with its core in The Netherlands and "arms" all over Europe.

A short observation at a single frequency from the core antennas is provided here. The m-file is a template that you should extend with your imaging algorithms.

3. Hints

Conduct a brief literature study to orient yourself on this problem. A suitable introduction to radio astronomy imaging is [1]. A LOFAR telescope dataset will be made available.

4. Consultants

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References

1. A.J. van der Veen and S. Wijnholds, "Signal processing tools for radio astronomy," in Handbook of Signal Processing Systems, 2nd ed. (S.S. Bhattacharryya e.a., ed.), pp. 421-464, Springer, May 2013. ISBN: 978-1-4614-6858-5 (Print); 978-1-4614-6859-2 (Online).